Towards an Involution Proof of Positivity for $Q_{n,c}(q)$: Connecting Commutation Relations, Local Rules, and the E–H Factorisation

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Abstract

We analyse the alternating-sign structure underlying the positivity conjecture for the cylindric Kostka polynomials $Q_{n,c}(q)$. By unpacking the E–H factorisation combinatorially, we identify the natural ground set for a sign-reversing involution: signed pairs (S, τ) where $S \subseteq \{1, \ldots, n\}$ carries the sign $(-1)^{|S|}$ and τ is a "purely cylindric" object counted by the nonneg polynomial $h_{n-|S|,c}(q)$. We argue that the local rules from the growth diagram framework provide the transformation mechanism for such an involution, exchanging subset decorations for cylindric structure while preserving q-weight.

Contents

1 The alternating-sign structure of $Q_{n,c}(q)$

1.1 The q-binomial decomposition

Recall that the polynomial $Q_{n,c}(q)$ is defined by

$$Q_{n,c}(q) := (q;q)_n \cdot [z^n]\Big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\Big) = (q;q)_n \cdot g_{n,c}(q),$$

where $g_{n,c}(q) = [z^n]\, G_c(z,q)$ and $G_c(z,q) = (zq;q)_\infty \cdot \mathrm{GK}_c(z,q)$ is the corrected generating function.

The q-binomial decomposition expresses $Q_{n,c}(q)$ as an alternating sum:

$$Q_{n,c}(q) = \sum_{j=0}^{n} (-1)^j\, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} \cdot h_{n-j,c}(q), \tag{1}$$

where $h_{m,c}(q) := (q;q)_m \cdot f_{m,c}(q)$ and $f_{m,c}(q) = [z^m]\, \mathrm{GK}_c(z,q)$ is the generating function for cylindric plane partitions with $\max = m$.

The sign $(-1)^j$ arises from the Euler expansion

$$(zq;q)_\infty = \prod_{k\geq 1}(1 - zq^k) = \sum_{j\geq 0} \frac{(-1)^j\, q^{j(j+1)/2}}{(q;q)_j}\, z^j.$$

Proving that $Q_{n,c}(q)$ has nonneg coefficients therefore requires showing that the alternating cancellations in (**??**) resolve perfectly into positivity.

1.2 Two sources of signs

Before the q-binomial simplification, $Q_{n,c}(q)$ arises from *two* sources of alternating signs:

1. The factor $(zq;q)_\infty = \prod_{k\geq 1}(1-zq^k)$ from the E–H factorisation, generating subsets $D \subset \{1,2,3,\ldots\}$ with sign $(-1)^{|D|}$.
2. The factor $(q;q)_n = \prod_{k=1}^{n}(1-q^k)$ from clearing denominators, generating subsets $T \subseteq \{1,\ldots,n\}$ with sign $(-1)^{|T|}$.

These two sign sources partially combine: the $(q;q)_n$ and $(q;q)_j$ merge into $\binom{n}{j}\cdot(q;q)_{n-j}$, and the second factor is absorbed into $h_{n-j,c}(q)$. In the resulting formula (**??**), the sign is carried entirely by $(-1)^j$ from the subset structure, while $h_{n-j,c}(q)$ contributes only positive terms.

2 The underlying combinatorial objects

2.1 Level A: Signed pairs from the E–H product

The E–H factorisation $G_c(z,q) = (zq;q)_\infty \cdot \mathrm{GK}_c(z,q)$ is a product of two generating functions, each with a combinatorial interpretation.

The Euler factor. $(zq;q)_\infty = \prod_{k\geq 1}(1-zq^k)$ generates finite subsets $D \subset \{1,2,3,\ldots\}$ (equivalently, partitions into distinct positive parts):

$$\prod_{k\geq 1}(1-zq^k) = \sum_{D\subset\{1,2,\ldots\}} (-1)^{|D|}\, z^{|D|}\, q^{\sigma(D)},$$

where $\sigma(D) = \sum_{d\in D} d$ is the sum of elements.

The CPP generating function. $\mathrm{GK}_c(z,q) = \sum_{\nu\in\mathrm{CPP}(c)} z^{\max(\nu)}\, q^{|\nu|}$ generates cylindric plane partitions of profile c, tracked by the max entry and total size.

The product. $G_c(z,q) = \sum_{(D,\,\nu)}(-1)^{|D|}\, z^{|D|+\max(\nu)}\, q^{\sigma(D)+|\nu|}$, where the sum ranges over all pairs (D,ν) with D a finite set of distinct positive integers and ν a CPP of profile c.

Extracting $[z^n]$ imposes the constraint $|D| + \max(\nu) = n$, so:

$$g_{n,c}(q) = \sum_{\substack{D\subset\{1,2,\ldots\},\ \nu\in\mathrm{CPP}(c)\\ |D|+\max(\nu)=n}} (-1)^{|D|}\, q^{\sigma(D)+|\nu|}. \tag{2}$$

Remark 1. At this level, the objects are **signed pairs** (D,ν): a distinct partition D and a cylindric plane partition ν whose max entries add up to n. The sign is determined by $|D|$ alone.

2.2 Level B: The clean form after clearing denominators

Multiplying by $(q;q)_n$ and using the q-binomial decomposition (**??**), the two sign sources merge into a single, cleaner description.

Definition 2 (Ground set for the involution)**.** For fixed $n \geq 0$ and profile $c = (c_0, c_1, c_2)$, define the ground set

$$\mathcal{G}_{n,c} := \bigsqcup_{j=0}^{n} \Big\{ (S, \tau) : S \subseteq \{1, \dots, n\},\ |S| = j,\ \tau \in \mathcal{H}_{n-j,\,c} \Big\},$$

where $\mathcal{H}_{m,c}$ is the (conjecturally nonempty for $m \geq 0$) set of combinatorial objects counted by $h_{m,c}(q)$, equipped with a weight function $\mathrm{wt}\colon \mathcal{H}_{m,c} \to \mathbb{N}$.

Each pair $(S, \tau) \in \mathcal{G}_{n,c}$ carries:

- **Sign:** $\varepsilon(S, \tau) := (-1)^{|S|}$.
- **Weight:** $w(S, \tau) := \sigma(S) + \mathrm{wt}(\tau)$, where $\sigma(S) = \sum_{s \in S} s$.

Remark 3 (Why subsets of $\{1, \dots, n\}$?)**.** The identity

$$q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S| = j}} q^{\sigma(S)}$$

gives the subset S its natural q-weight. This follows because a j-element subset $\{s_1 < \cdots < s_j\} \subseteq \{1, \dots, n\}$ bijects to a partition λ fitting in a $j \times (n-j)$ box via $\lambda_i = s_i - i$ (after reindexing), and $\sigma(S) = |\lambda| + j(j+1)/2$.

With this definition, the q-binomial decomposition becomes:

$$Q_{n,c}(q) = \sum_{(S,\tau) \in \mathcal{G}_{n,c}} (-1)^{|S|}\, q^{\sigma(S) + \mathrm{wt}(\tau)}. \tag{3}$$

Positivity of $Q_{n,c}(q)$ is therefore equivalent to the existence of a sign-reversing, weight-preserving involution on $\mathcal{G}_{n,c}$ whose fixed points all have $|S|$ even (positive sign).

3 What does $h_{m,c}(q)$ count?

3.1 The definition

Recall

$$h_{m,c}(q) = (q;q)_m \cdot f_{m,c}(q),$$

where $f_{m,c}(q) = [z^m]\ \mathrm{GK}_c(z, q) = \sum_{\nu\colon \max(\nu) = m} q^{|\nu|}$.

Since $1/(q;q)_m$ is the generating function for partitions of length $\leq m$ (equivalently, partitions with largest part $\leq m$ after conjugation), the multiplication by $(q;q)_m$ "cancels the partition overhead": it removes the contribution of the base partition γ in the growth diagram.

3.2 Growth diagram interpretation

The growth diagram bijection $\psi\colon (\mathrm{IP}, \mathrm{ALCD}(c)) \to \mathrm{CPP}(c)$ decomposes each CPP into a base partition $\gamma \in \mathrm{IP}$ and an arbitrarily labelled cylindric diagram $d \in \mathrm{ALCD}(c)$. The generating function factors as:

$$\mathrm{GK}_c(z, q) = \sum_{(\gamma,\, d)} z^{\max(\psi(\gamma,d))}\, q^{|\psi(\gamma,d)|}.$$

The E–H factorisation $\mathrm{GK}_c(z, q) = H(z) \cdot G_c(z, q)$ corresponds, at the level of generating functions, to separating:

- The contribution of γ (an ordinary partition), which gives the universal factor $H(z) = 1/(zq;q)_\infty = \sum_{n\geq 0} z^n/(q;q)_n$.
- The "purely cylindric" content, which gives $G_c(z,q)$.

Remark 4. The objects $\tau \in \mathcal{H}_{m,c}$ counted by $h_{m,c}(q)$ should therefore be understood as **equivalence classes of ALCDs** (or a related combinatorial species) that encode the purely cylindric structure at max level m, with the ordinary-partition freedom of the base partition γ factored out.

Computational evidence confirms that $h_{m,c}(q)$ is a polynomial with nonneg integer coefficients for all tested profiles — a strictly stronger observation than positivity of $Q_{n,c}(q)$ itself, since $Q_{n,c}(q)$ is built from the $h_{m,c}(q)$ via the alternating-sign sum (**??**).

4 The involution mechanism via local rules

4.1 Requirements for the involution

Definition 5 (Sign-reversing involution)**.** A *sign-reversing, weight-preserving involution* on $\mathcal{G}_{n,c}$ is a map $\iota\colon \mathcal{G}_{n,c} \to \mathcal{G}_{n,c}$ satisfying:

1. $\iota^2 = \mathrm{id}$ (involution).
2. If $\iota(S,\tau) = (S',\tau') \neq (S,\tau)$, then $|S'| \equiv |S| + 1 \pmod 2$ (sign reversal).
3. $\sigma(S) + \mathrm{wt}(\tau) = \sigma(S') + \mathrm{wt}(\tau')$ (weight preservation).
4. Every fixed point $(S,\tau) = \iota(S,\tau)$ satisfies $|S| \equiv 0 \pmod 2$ (positive sign).

The existence of such an involution is equivalent to the positivity of $Q_{n,c}(q)$. The fixed-point generating function gives:

$$Q_{n,c}(q) = \sum_{\substack{(S,\tau)\in\mathcal{G}_{n,c}\\ \iota(S,\tau)=(S,\tau)}} q^{\sigma(S)+\mathrm{wt}(\tau)}.$$

4.2 How the involution should act

The involution must simultaneously manipulate two different structures:

1. The **subset** $S \subseteq \{1,\dots,n\}$ (combinatorially simple).
2. The **cylindric object** $\tau \in \mathcal{H}_{m,c}$ (governed by interlacing conditions encoding the A_2 root system).

Concretely, the involution trades between these two worlds:

Strategy 6 (Peel/glue mechanism)**.** Given a non-fixed pair (S,τ) with $\tau \in \mathcal{H}_{m,c}$ and $m = n - |S|$:

Peel step ($|S| \to |S|+1$, reducing m by 1):

1. Identify a canonical element $k \in \{1,\dots,n\} \setminus S$, determined by the structure of τ.
2. "Peel off" a strip of weight k from τ using an inverse local rule, obtaining $\tau' \in \mathcal{H}_{m-1,c}$ with $\mathrm{wt}(\tau') = \mathrm{wt}(\tau) - k$.
3. Set $S' = S \cup \{k\}$.

4. The new pair (S', τ') has sign $(-1)^{|S|+1}$ and weight $\sigma(S')+\mathrm{wt}(\tau') = (\sigma(S)+k)+(\mathrm{wt}(\tau)-k) = \sigma(S) + \mathrm{wt}(\tau)$.

Glue step ($|S| \to |S|-1$, increasing m by 1):

1. Identify a canonical element $k \in S$, determined by the structure of τ.
2. "Glue on" a strip of weight k to τ using a forward local rule, obtaining $\tau' \in \mathcal{H}_{m+1,c}$ with $\mathrm{wt}(\tau') = \mathrm{wt}(\tau) + k$.
3. Set $S' = S \setminus \{k\}$.
4. Weight is preserved by the same calculation.

Remark 7. The decision of whether to peel or glue (and the choice of k) must be **canonical** — determined entirely by (S, τ) — and must satisfy $\iota^2 = \mathrm{id}$. The fixed points are the pairs where neither peeling nor gluing is possible: the "irreducible" configurations.

4.3 Local rules as the transformation engine

The growth diagram framework provides the mechanism for the peel/glue operations. Recall the algebra–bijection dictionary:

Algebraic operation	**Bijective operation**
Commuting $\Omega^*[Xu]$ past $\Omega[Xv]$	Applying local rule $\mathfrak{L}_i$
Scalar factor $\frac{1}{1-uv} = \sum_{m\geq 0}(uv)^m$	Face label $m \geq 0$
Inverse: $\Omega[Xv]\,\Omega^*[Xu] \to \Omega^*\,\Omega \cdot (1-uv)$	Inverse local rule: absorb face label

Each application of the local rule $\mathfrak{L}_i$ transforms one face of the growth diagram. The local rule has type signature

$$\mathfrak{U}_{\alpha,\beta} \colon U(\alpha) \cap U(\beta) \to (\mathbb{Z}_{\geq 0},\ D(\alpha) \cap D(\beta)),$$

and satisfies the weight condition $|\lambda| = |\alpha| + |\beta| - |\mu| + m$, where m is the face label.

Forward local rule (glue): Given μ, α, β and a face label $m \geq 0$, determine λ. This *increases* the partition sizes (adds a strip).

Inverse local rule (peel): Given λ, α, β, determine μ and the face label m. This *decreases* the partition sizes (removes a strip) and *extracts* the face label m.

The peel step of the involution extracts a face label m from the cylindric object τ (reducing its weight by m) and deposits the value m into the subset S. The glue step does the reverse.

4.4 Analogy with Franklin's involution

The proposed structure is closely analogous to **Franklin's involution** for Euler's pentagonal number theorem.

In Franklin's setting, the identity $\prod_{k\geq 1}(1-q^k) = \sum_{m\in\mathbb{Z}}(-1)^m\, q^{m(3m-1)/2}$ is proved by a sign-reversing involution on partitions into distinct parts. The involution moves the smallest part to/from the "Durfee edge," and fails (producing fixed points) precisely at the pentagonal numbers.

In our setting:

Franklin's involution	**Proposed involution**
Partitions into distinct parts	Pairs (S, τ): subset + cylindric object
Move smallest part $\leftrightarrow$ Durfee edge	Peel/glue strip via local rule
Sign from $(-1)^{\#\text{parts}}$	Sign from $(-1)^{\lvert S\rvert}$
Fixed points: pentagonal configurations	Fixed points: irreducible (S, τ)
Pentagonal number theorem	Positivity of $Q_{n,c}(q)$

The essential new difficulty is that Franklin's involution acts within a single combinatorial species (distinct partitions), whereas our involution must bridge two worlds: the subset structure and the cylindric partition structure, mediated by the local rules.

5 A concrete example

5.1 The case $d = 4$, $c = (2, 2, 0)$, $n = 1$

For this case, $Q_{1,(2,2,0)}(q) = q^2 + q^3 + q^4 + q^5$. The q-binomial decomposition (**??**) gives:

$$Q_{1,c}(q) = h_{1,c}(q) - q \cdot h_{0,c}(q).$$

The h-polynomials.

- $h_{0,c}(q) = (q;q)_0 \cdot f_{0,c}(q) = 1$ (the empty CPP).
- $h_{1,c}(q) = (1-q) \cdot f_{1,c}(q)$. Since $Q_{1,c} = h_{1,c} - q$ and $Q_{1,c} = q^2 + q^3 + q^4 + q^5$, we get $h_{1,c}(q) = q + q^2 + q^3 + q^4 + q^5$.

The ground set $\mathcal{G}_{1,c}$.

- $j = 0$ **(sign +):** Pairs $(\emptyset, \tau)$ with $\tau \in \mathcal{H}_{1,c}$. Five objects at weights $1, 2, 3, 4, 5$.
- $j = 1$ **(sign −):** Pairs $(\{1\}, \tau_0)$ with $\tau_0 \in \mathcal{H}_{0,c}$. One object at weight $\sigma(\{1\}) + 0 = 1$.

The involution. The unique negative object $(\{1\}, \tau_0)$ at weight 1 pairs with the unique positive object $(\emptyset, \tau_1^{\min})$ at weight 1, where $\tau_1^{\min}$ is the minimal-weight element of $\mathcal{H}_{1,c}$. The pairing says: the minimal-weight cylindric object at max level 1 can be "peeled" (via an inverse local rule extracting a strip of weight 1) to obtain the empty cylindric object at max level 0.

The four surviving fixed points at weights $2, 3, 4, 5$ give $Q_{1,c}(q) = q^2 + q^3 + q^4 + q^5$.

5.2 Interpretation

In this example, the involution is uniquely forced: there is only one negative object and only one positive object at the matching weight. For larger n, multiple pairings become possible and the *choice of canonical pairing* — determined by the internal structure of the cylindric object τ — becomes the central difficulty.

6 The fundamental difficulties

6.1 Bridging two worlds

The involution must operate across two fundamentally different combinatorial domains:

1. **The subset world** ($S \subseteq \{1, \ldots, n\}$): well-understood, governed by elementary combinatorics.

2. **The cylindric world** ($\tau \in \mathcal{H}_{m,c}$): governed by the interlacing conditions $\nu_j^{(i)} \geq \nu_{j+c_{i+1}}^{(i+1)}$ that encode the A_2 root system.

The local rules provide a *mechanism* for transforming cylindric objects (each rule application corresponds to one commutation step), but designing a *canonical* choice — which element k to peel or glue, determined entirely by (S, τ), satisfying $\iota^2 = \mathrm{id}$ — requires detailed understanding of the cylindric combinatorics.

6.2 Relation to the missing charge statistic

The fixed points of the involution would simultaneously solve the "missing charge" problem:

- The fixed-point set $\mathrm{Fix}(\iota) \subset \mathcal{G}_{n,c}$ has generating function $Q_{n,c}(q)$.
- Each fixed point (S, τ) has $|S|$ even, and its weight $\sigma(S) + \mathrm{wt}(\tau)$ defines the "charge" statistic.
- The cardinality $|\mathrm{Fix}(\iota)| = Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$.

So constructing the involution is equivalent to finding the charge statistic: the fixed points *are* the charged objects.

6.3 Relation to the cyclic descent injection

The cyclic descent injection (previously established for $d \not\equiv 0 \pmod 3$) provides a weight-raising map *within* a single max level: it injects weight-$(w-1)$ CPPs with max 1 into weight-w CPPs with max 1. This addresses the monotonicity of coefficients of $h_{1,c}(q)$.

The involution proposed here operates *across* max levels, trading cylindric structure for subset decorations. These are complementary:

- The cyclic descent injection establishes structure *within* each $\mathcal{H}_{m,c}$.
- The involution establishes cancellation *between* the $\mathcal{H}_{m,c}$ for different m, mediated by the subsets S.

A complete proof would likely require both ingredients.

7 Outlook: what is needed

To implement the involution strategy, the following steps are needed:

1. **Concrete model for $\mathcal{H}_{m,c}$.** Give an explicit combinatorial description of the objects counted by $h_{m,c}(q)$. The growth diagram suggests these are ALCDs with the base-partition freedom factored out, but a precise bijective description is needed.

2. **Canonical peel/glue rule.** Define a deterministic rule that, given (S, τ), either:
   - selects $k \in \{1, \ldots, n\} \setminus S$ and peels a strip of weight k from τ, or
   - selects $k \in S$ and glues a strip of weight k onto τ, or

- declares (S, τ) a fixed point.

The rule must be involutive: applying peel then glue (or vice versa) returns to the original pair.

3. **Identification of fixed points.** Characterise the irreducible pairs (S, τ) that are fixed by the involution. These should form a set of cardinality $Q_{n,c}(1) = \left(\frac{(d+1)(d+2)}{6} - 1\right)^n$ and their weight distribution should give $Q_{n,c}(q)$.

4. **Verification.** Test the proposed involution computationally for small cases ($d = 4$, $n \leq 3$) before attempting a general proof.

The local rules from the growth diagram framework — the bijective incarnation of the commutation relations — are the natural candidates for the peel/glue mechanism. The challenge is to harness them in a way that produces a globally consistent involution.